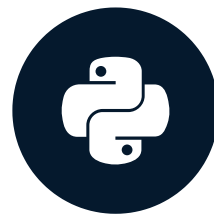


Adding time to the mix

WORKING WITH DATES AND TIMES IN PYTHON



Max Shron

Data Scientist and Author

Dates and Times

Date
October 1 2017,

Time
3: 23: 25 PM

Dates and Times

Date
October 1 2017,

Time
3: 23: 25 PM

```
# Import datetime  
from datetime import datetime
```

Dates and Times

Date
October 1 2017,

Time
3: 23: 25 PM

```
# Import datetime
from datetime import datetime

dt = datetime(
```

Dates and Times

Date
October 1 2017,

Time
3: 23: 25 PM

```
# Import datetime  
from datetime import datetime  
  
dt = datetime(2017, 10, 1
```

Dates and Times

Date
October 1 2017,

Time
3: 23: 25 PM

```
# Import datetime
from datetime import datetime

dt = datetime(2017, 10, 1, 15
```

Dates and Times

Date
October 1 2017,

Time
3: 23: 25 PM

```
# Import datetime
from datetime import datetime

dt = datetime(2017, 10, 1, 15, 23,
```

Dates and Times

Date
October 1 2017,

Time
3: 23: 25 PM

```
# Import datetime
from datetime import datetime

dt = datetime(2017, 10, 1, 15, 23, 25)
```

Dates and Times

Date
October 1 2017,

Time
3: 23: 25 PM

```
# Import datetime
from datetime import datetime

dt = datetime(2017, 10, 1, 15, 23, 25, 500000)
```

Dates and Times

Date
October 1 2017,

Time
3: 23: 25 PM

```
# Import datetime
from datetime import datetime

dt = datetime(year=2017, month=10, day=1,
              hour=15, minute=23, second=25,
              microsecond=500000)
```

Replacing parts of a datetime

```
print(dt)
```

```
2017-10-01 15:23:25.500000
```

```
dt_hr = dt.replace(minute=0, second=0, microsecond=0)  
print(dt_hr)
```

```
2017-10-01 15:00:00
```

Capital Bikeshare



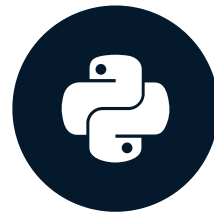
Capital Bikeshare Station Installed at the Lincoln Memorial by Euan Fisk, licensed CC B 2.0

Adding time to the mix

WORKING WITH DATES AND TIMES IN PYTHON

Printing and parsing datetimes

WORKING WITH DATES AND TIMES IN PYTHON



Max Shron

Data Scientist and Author

Printing datetimes

```
# Create datetime  
dt = datetime(2017, 12, 30, 15, 19, 13)  
print(dt.strftime("%Y-%m-%d"))
```

```
2017-12-30
```

```
print(dt.strftime("%Y-%m-%d %H:%M:%S"))
```

```
2017-12-30 15:19:13
```

Printing datetimes

```
print(dt.strftime("%H:%M:%S on %Y/%m/%d/"))
```

```
15:19:13 on 2017/12/30
```

ISO 8601 Format

```
# ISO 8601 format  
print(dt.isoformat())
```

```
2017-12-30T15:19:13
```

Parsing datetimes with strptime

```
# Import datetime  
from datetime import datetime
```

Parsing datetimes with strptime

```
# Import datetime
from datetime import datetime

dt = datetime.strptime(
```

Parsing datetimes with strptime

```
# Import datetime
from datetime import datetime

dt = datetime.strptime("12/30/2017 15:19:13"
```

Parsing datetimes with strptime

```
# Import datetime
from datetime import datetime

dt = datetime.strptime("12/30/2017 15:19:13",
                       "%m/%d/%Y %H:%M:%S")
```

Parsing datetimes with.strptime

```
# What did we make?  
print(type(dt))
```

```
<class 'datetime.datetime'>
```

```
# Print out datetime object  
print(dt)
```

```
2017-12-30 15:19:13
```

Parsing datetimes with strptime

```
# Import datetime
from datetime import datetime

# Incorrect format string
dt = datetime.strptime("2017-12-30 15:19:13", "%Y-%m-%d")
```

```
ValueError: unconverted data remains: 15:19:13
```

Parsing datetimes with Python

```
# A timestamp  
ts = 1514665153.0  
  
# Convert to datetime and print  
print(datetime.fromtimestamp(ts))
```

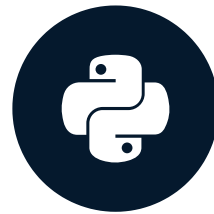
```
2017-12-30 15:19:13
```

Printing and parsing datetimes

WORKING WITH DATES AND TIMES IN PYTHON

Working with durations

WORKING WITH DATES AND TIMES IN PYTHON



Max Shron

Data Scientist and Author

Working with durations



Working with durations



```
# Create example datetimes
```

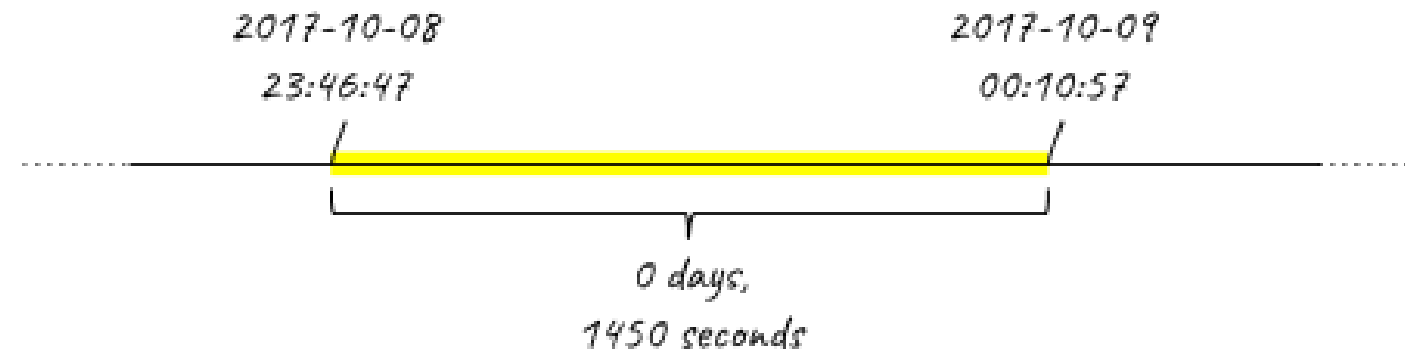
```
start = datetime(2017, 10, 8, 23, 46, 47)
```

```
end = datetime(2017, 10, 9, 0, 10, 57)
```

```
# Subtract datetimes to create a timedelta
```

```
duration = end - start
```

Working with durations



```
# Subtract datetimes to create a timedelta  
print(duration.total_seconds())
```

```
1450.0
```

Creating timedeltas

```
# Import timedelta  
from datetime import timedelta
```

```
# Create a timedelta  
delta1 = timedelta(seconds=1)
```

Creating timedeltas

```
print(start)
```

```
2017-10-08 23:46:47
```

```
# One second later  
print(start + delta1)
```

```
2017-10-08 23:46:48
```

Creating timedeltas

```
# Create a one day and one second timedelta  
delta2 = timedelta(days=1, seconds=1)
```

```
print(start)
```

```
2017-10-08 23:46:47
```

```
# One day and one second later  
print(start + delta2)
```

```
2017-10-09 23:46:48
```

Negative timedeltas

```
# Create a negative timedelta of one week  
delta3 = timedelta(weeks=-1)
```

```
print(start)
```

```
2017-10-08 23:46:47
```

```
# One week earlier  
print(start + delta3)
```

```
2017-10-01 23:46:47
```

Negative timedeltas

```
# Same, but we'll subtract this time  
delta4 = timedelta(weeks=1)
```

```
print(start)
```

```
2017-10-08 23:46:47
```

```
# One week earlier  
print(start - delta4)
```

```
2017-10-01 23:46:47
```

Working with durations

WORKING WITH DATES AND TIMES IN PYTHON